



MELANTERITE
Monoclinic
H 2 • SG 1.89

White, green, or blue melanterite is hydrated iron sulfate. It is used in purifying water supplies, and as a fertilizer.



JAROSITE
Trigonal

H $2\frac{1}{2}$ – $3\frac{1}{2}$ • SG 2.9–3.26

Jarosite is a hydrous sulfate of iron and potassium, occurring as brown coatings on pyrite and other iron minerals.

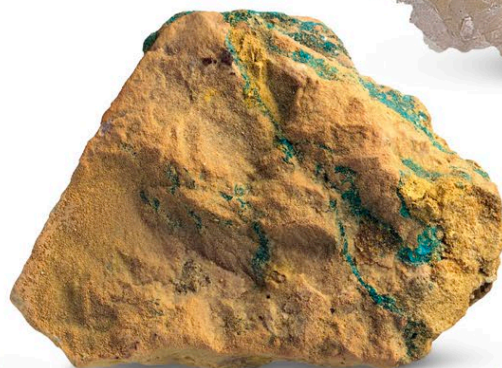


CELESTINE

Orthorhombic • H $3\frac{1}{2}$ • SG 3.96–3.98

This strontium sulfate is sought after, not only as the main source of strontium, but also for the beautiful, transparent, pale-colored crystals.

EPSOMITE
Orthorhombic
H $2\frac{1}{2}$ • SG 1.68
This hydrated magnesium sulfate occurs in arid regions and limestone cave walls. It is the source of the laxative Epsom salts.



COPIAPITE

Triclinic • H $2\frac{1}{2}$ –3 • SG 2.08–2.17

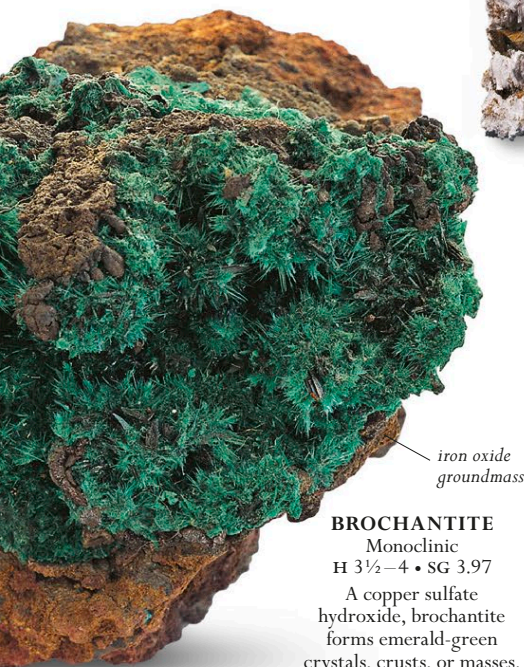
A yellow or green hydrated sulfate of iron first described from Copiapó, Chile, copiapite occurs where other minerals are altered.



BARITE

Orthorhombic
H 3 • SG 4.5

The most common barium mineral, barium sulfate is unusually heavy for a pale-colored mineral.



BROCHANTITE

Monoclinic
H $3\frac{1}{2}$ –4 • SG 3.97

A copper sulfate hydroxide, brochantite forms emerald-green crystals, crusts, or masses.



ANHYDRITE

Orthorhombic
H $3\frac{1}{2}$ • SG 2.98

A form of calcium sulfate, anhydrite occurs with gypsum but is less common. It alters to gypsum in humid conditions.



POLYHALITE

Triclinic • H $2\frac{1}{2}$ – $3\frac{1}{2}$ • SG 2.78

Polyhalite is a hydrated potassium calcium magnesium sulfate. Colourless, white, pink, or red, it is widespread in many marine salt deposits.

MOLYBDATES

Molybdates form when metals combine with the molybdate radical (MoO_4). These minerals are rare and tend to be dense and brightly coloured. Molybdate minerals occur in mineral veins that have been altered by circulating water. Wulfenite is the best-known molybdate mineral. It is prized for its fine crystals and brilliant orange or yellow colours.



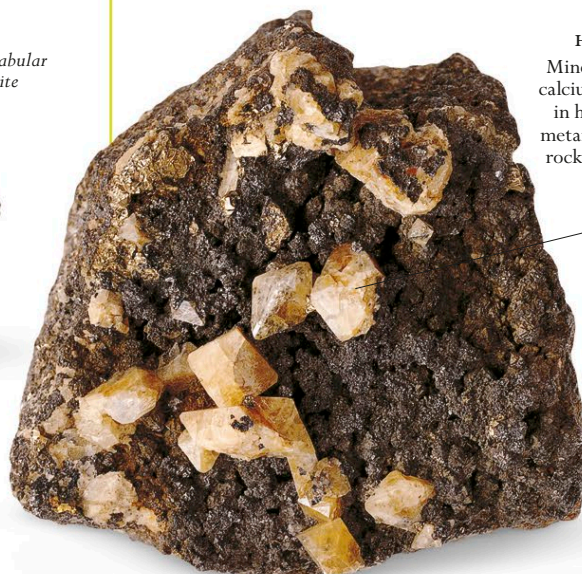
WULFENITE

Tetragonal • H $2\frac{1}{2}$ –3 • SG 6.5–7.5

This lead molybdate is found in oxidized zones of lead and molybdenum deposits. It is a minor source of molybdenum.

TUNGSTATES

Tungstate minerals are compounds with metallic elements joined to the tungstate radical (WO_4). These minerals are rare and usually have a high specific gravity. Some form as fine crystals. Tungstates occur in hydrothermal veins and pegmatites – very coarse-grained granitic rocks where minerals form from fluids permeating the rock.



SCHEELITE

Tetragonal
H $4\frac{1}{2}$ –5 • SG 6.1

Mined for tungsten, this calcium tungstate is found in hydrothermal veins, metamorphic and igneous rocks, and alluvial sands.

bipyramidal
scheelite crystal

HÜBNERITE

Monoclinic
H $4\frac{1}{2}$ –5 • SG 7.12–7.18

This manganese tungstate is a major source of tungsten, a metal used in steel alloys, abrasives, and light bulbs.



quartz
hübnerite
crystal

FERBERITE

Monoclinic
H $4\frac{1}{2}$ –5 • SG 7.58

Opaque black ferberite is found in hydrothermal veins and granitic pegmatites. It is an iron tungstate, mined as a source of tungsten.

